

## Reference excerpt 01-040

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**Pressure gradient calculation** The pressure inside the IJV is calculated from the instantaneous CSA as follows:  $p(t) = 1/C \cdot CSA(t)$  (6) substituting the expression of  $p(t)$  obtained above into Eq. (2) we obtain the expression for the pressure gradient:  $p_z = 1/C \cdot dCSA/dt$  (7)

**Flow velocity calculation**  $w(t, z)$  The function  $w(t, z)$  is calculated by inserting into the Eq.(1) the expression for the pressure gradient found in Eq. (7). The mathematical details are given in [Sisini et al.(2015b)].

**Figure 3:** Calculated velocity trace together with the experimental one obtained by the Doppler examination.

**RESULTS EXAMPLE**

**Pressure waves velocity calculation** The measured delay ( $\tau_{QP}$ ) was 0.14 s and the distance IG was 23 cm. As a consequence the velocity  $c$  was 164 cm/s.

**Compliance calculation** The minimum value of the CSA ( $CSA_x$ ) during the cardiac cycle was 0.2 cm<sup>2</sup> and the compliance for unit of length was 9.8 10<sup>-3</sup> cm<sup>2</sup>/mmHg.

**Flow velocity calculation**  $w(t, z)$  The velocity was calculated following Eq. (1) and its trace is shown in Fig. 3 together with the experimental trace obtained by the Doppler examination. The two traces in agreement, nevertheless, since the velocity  $w(t, G + l)$  was calculated up to an additive constant, only the wave amplitude has physical meaning whereas its shift along the velocity axis is not significant.

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